Laboratory Project: Logistic Sequences

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Introduction

- This presentation explores how a simple formula (Logistic Sequence that models a population) can behave in surprising ways. By changing the starting value and growth rate, we'll see how the stable outcomes change to unpredictable, chaotic patterns. Using Python to run the computations and generate graphs, we'll break down different cases to better understand how small changes can lead to big differences in results.

Definition

The logistic sequence is defined as:

$$p_{n+1} = k\, p_n\, (1 - p_n)$$

- p_n: normalized population at step n, $0 < p_n < 1$
- k: growth rate
- p_0: initial population

Behavior Types

Depending on p_0 and k, the sequence can show:

- **Convergent behavior**: settles to a fixed point
- **Periodic behavior**: oscillates in a cycle
- **Chaotic behavior**: highly sensitive and unpredictable

These behaviors illustrate how even a simple formula (model) can produce complex and varied population dynamics observed in nature

Notes and References

- We limit $p_0 \in (0, 1)$ for simplicity.
- For mathematical background, see:
  - Milefoot
  - Wikipedia
  - James Stewart. Calculus: Early Transcendentals

Methodology

- Run Python code for many (p_0, k) combinations
- Iterate each sequence for $n = 100$ steps
- Save all data to a CSV file
- Automate plotting of each sequence as a graph

Python: Generating Sequences

```
import csv
n = 100
p_values = [0.6572, 0.5, 0.892]
k_values = [1.964, 1.2345, 2.7852]

with open("runs.csv", 'w', newline='') as file:
    writer = csv.writer(file)
    header = [f"p={p}_k={k}" for p in p_values for k in
        k_values]
    writer.writerow(header)

    sequences = []
    for p0 in p_values:
        for k in k_values:
            p = p0
            sequence = []
            for i in range(n):
                sequence.append(p)
                p = k * p * (1 - p)
            sequences.append(sequence)

    for i in range(n):
        writer.writerow([seq[i] for seq in sequences])
```

CSV Structure

Each column corresponds to one (p_0, k) pair.

Column structure:

- Row 1: header - parameters used
- Row 2: initial value p_0
- Remaining rows: values after each iteration

p=0.6572, k=1.96	p=0.6572, k=1.23	p=0.6572, k=2.78
0.6572	0.6572	0.6572
0.4425	0.2781	0.6274
0.4846	0.2478	0.6510

Plotting the Sequences

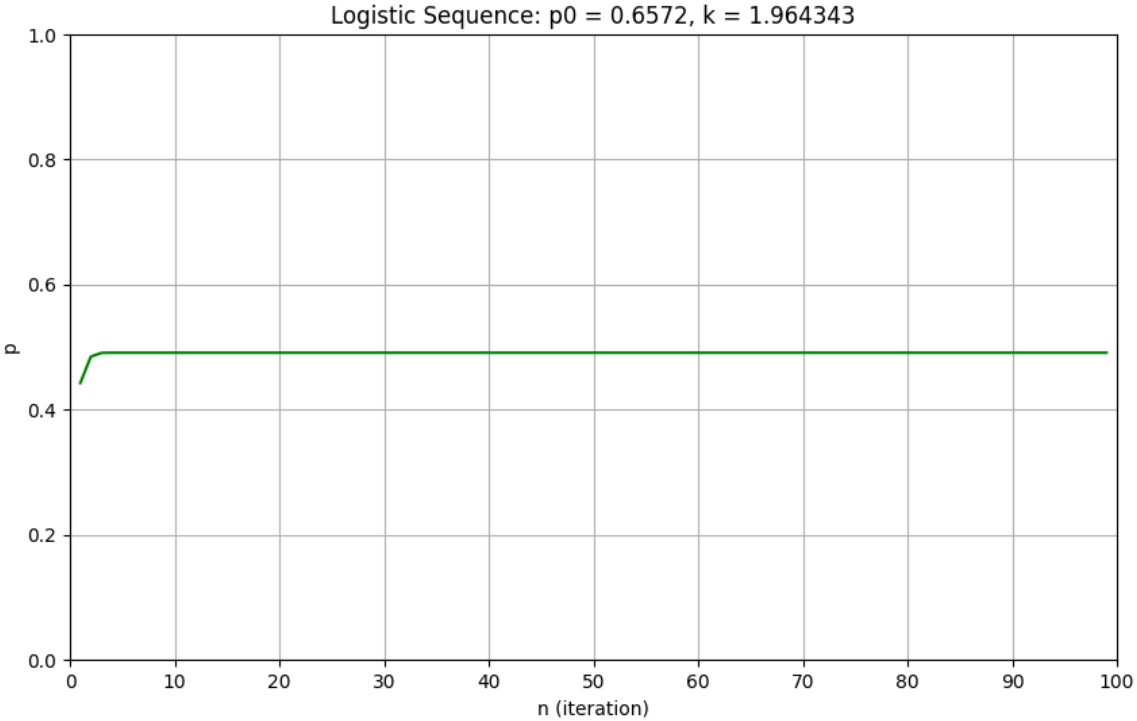
A second Python program reads the CSV and generates one graph per run.

For each plot:

- X-axis: iteration (1–100)
- Y-axis: population value p_n
- Each graph shows one run
- Saved as PNG images

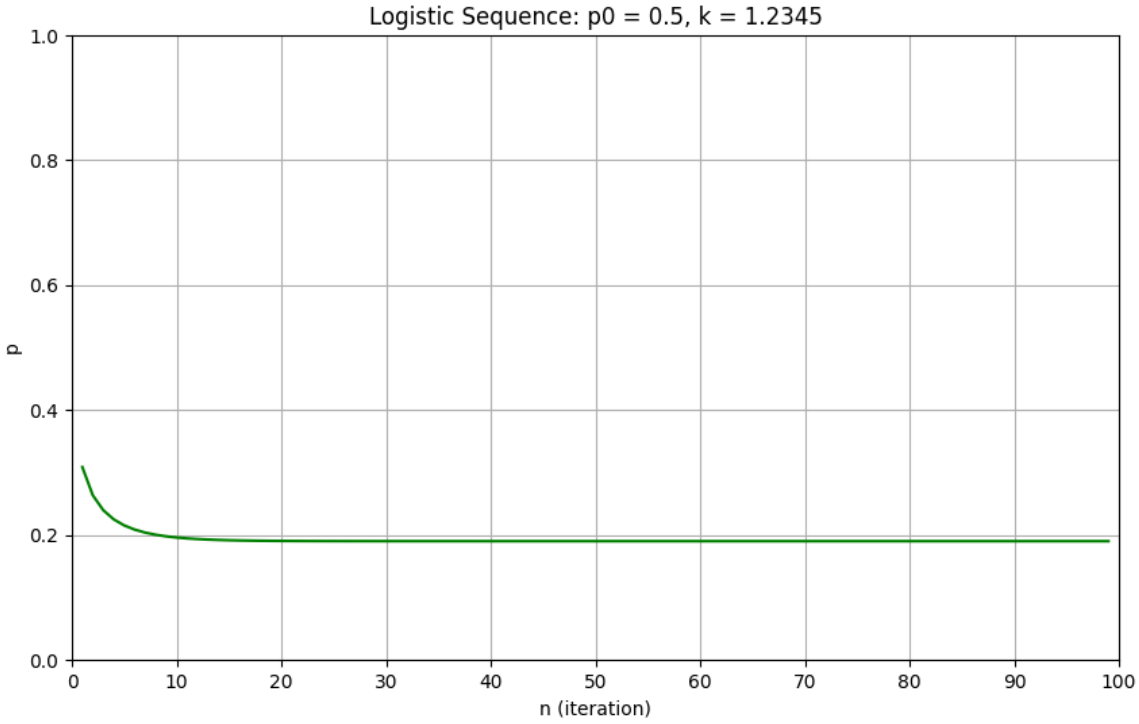
See full code on Drive

Sample Graph



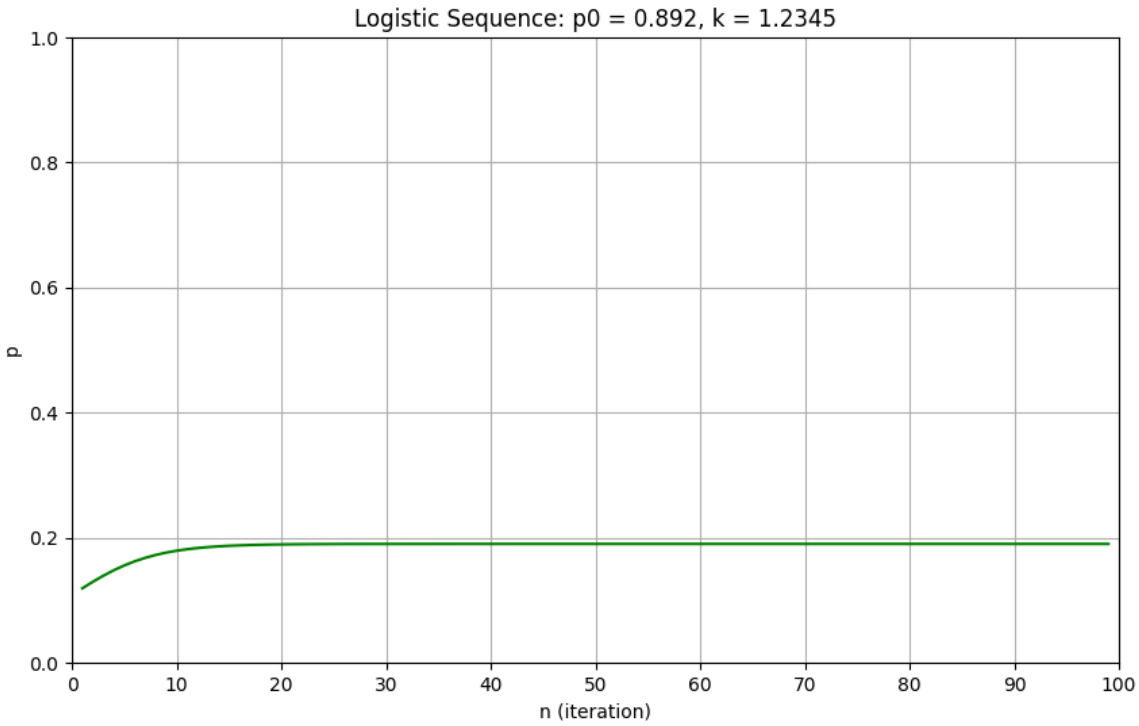
Experiment1: $1 < k < 3$

- Example: $k = 1.2345$, $p_0 = 0.5$
- The sequence converges quickly to approximately 0.2, indicating a stable behavior.



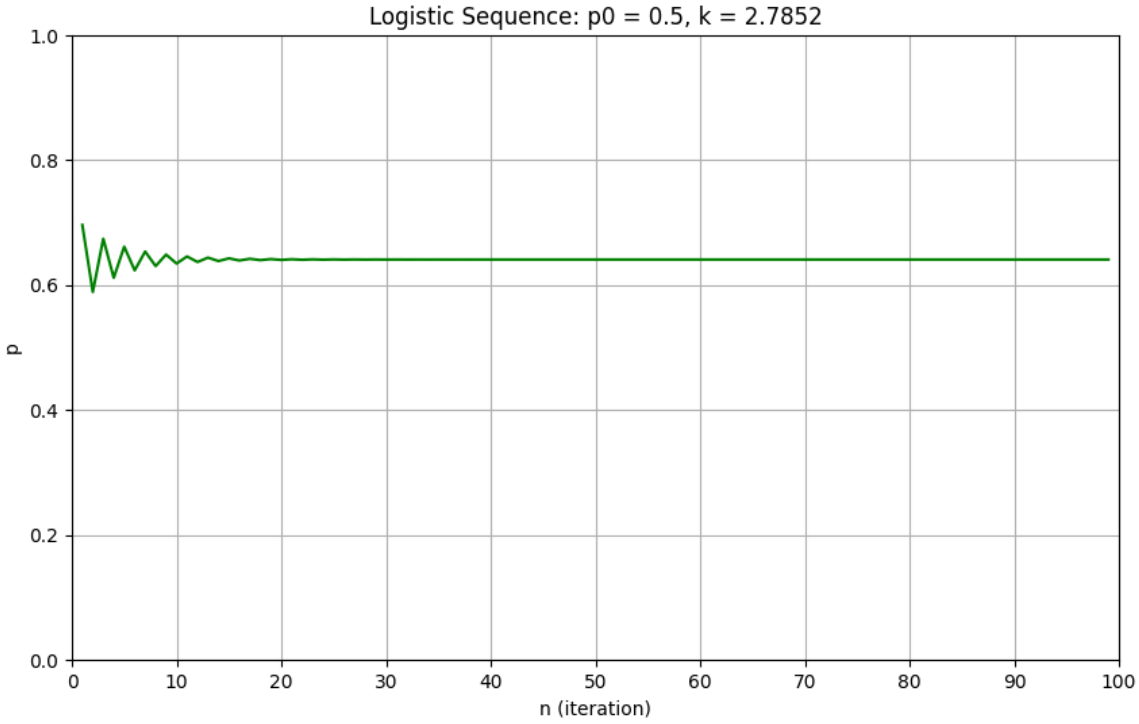
Experiment1: $1 < k < 3$

- Same $k = 1.2345$, different $p_0 = 0.892$
- Converges to same point



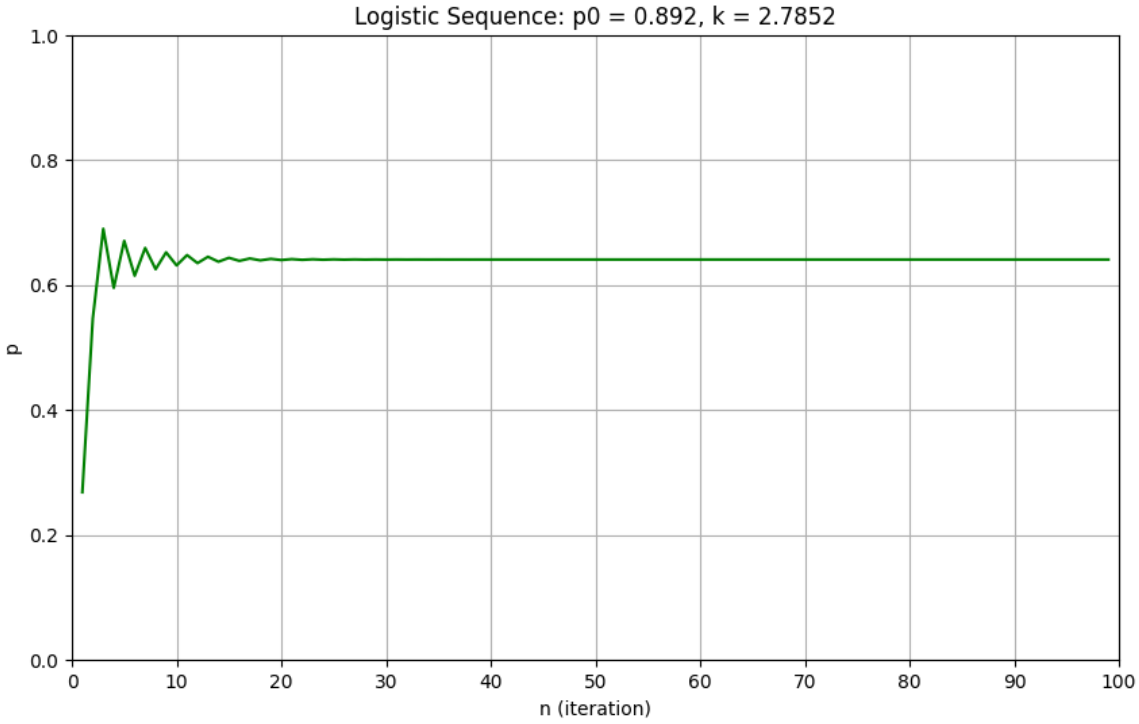
Experiment1: $1 < k < 3$

- Example: k $= 2.7852$, $p_0 = 0.5$
- The sequence converges to around 0.6, again suggesting a stable limit point



Experiment1: $1 < k < 3$

- Same $k = 2.7852$, different $p_0 = 0.892$
- The sequence converges to about 0.6, like before, further confirming this pattern.

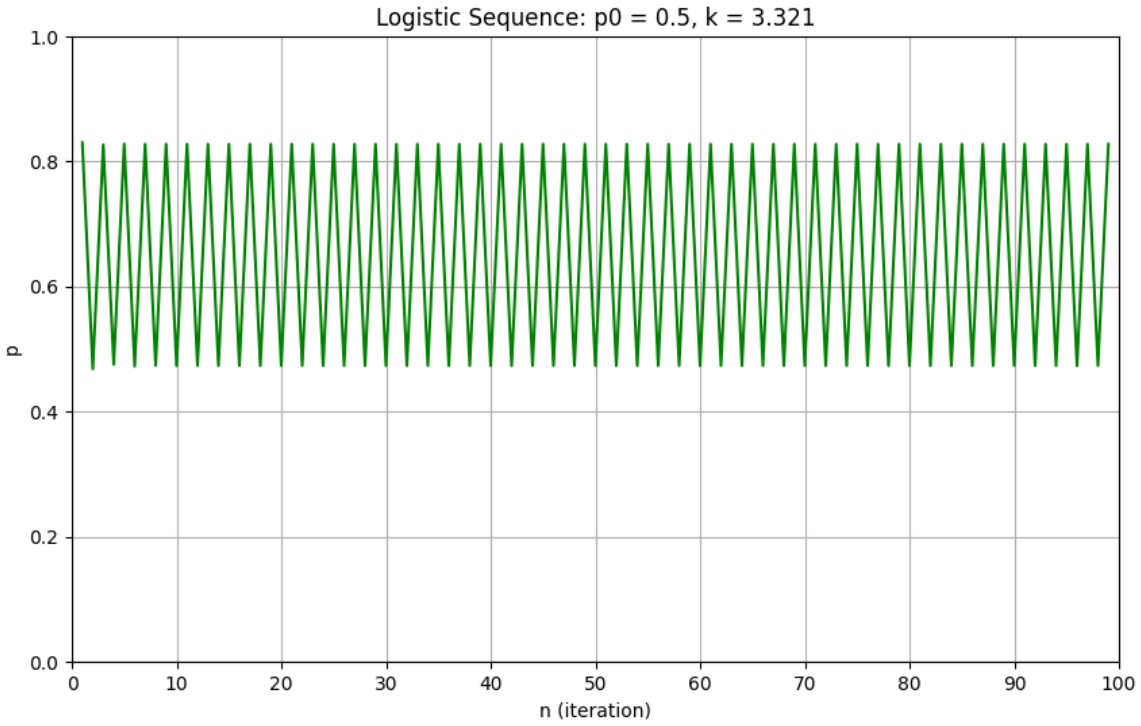


Experiment1 - Observations

- The limit of the sequence **does not significantly depend on the initial value** p_0 (as long as it is in the interval $(0, 1)$).
- The **growth rate** k has a greater influence on the value to which the sequence converges.
- For $k \in (1, 3)$, the logistic sequence exhibits **convergent behavior**, settling to a stable fixed point.

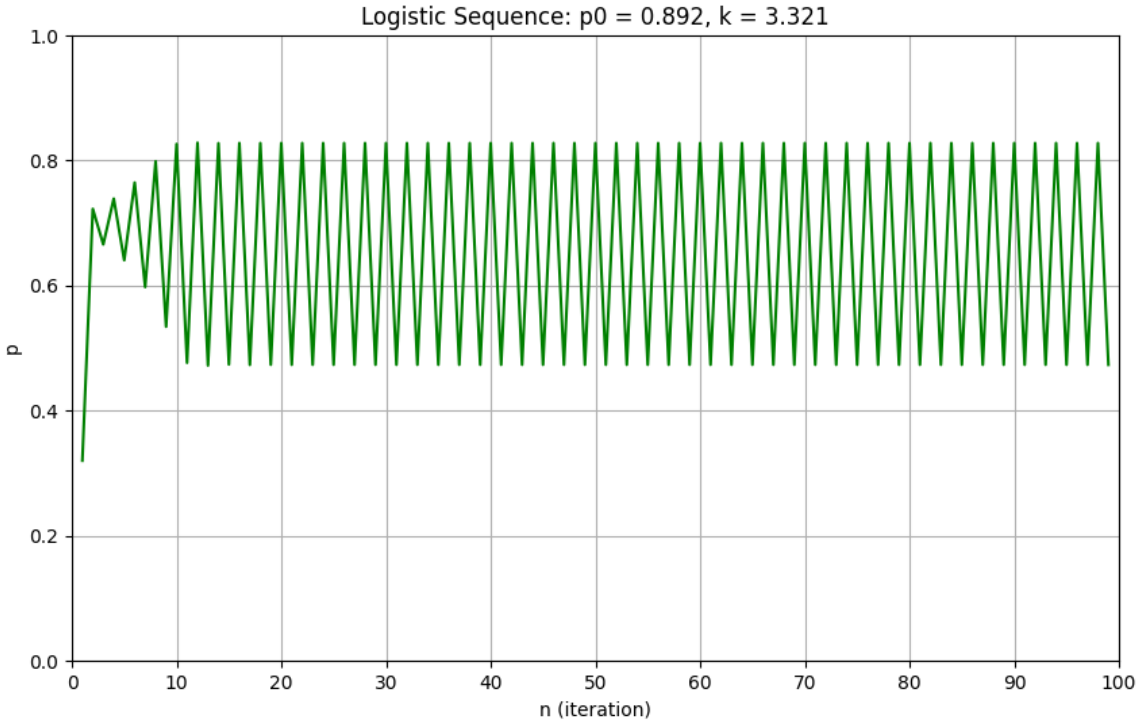
Experiment2: $3 < k < 3.4$

- For this case, the parameter k was set to 3.321
- For p0 = 0.5, the graph exhibits a periodic oscillation right from the beginning, with the sequence fluctuating between approximately 0.5 and 0.8 in a consistent manner. This suggests the sequence has entered a stable cycle of values after very few iterations



Experiment2: $3 < k < 3.4$

- For this case, the parameter k was set to 3.321
- For p0 = 0.892, the behavior is similar, although the periodic oscillation becomes visible only after around 10 iterations. Once stabilized, the sequence also oscillates evenly between similar values as in the first case.

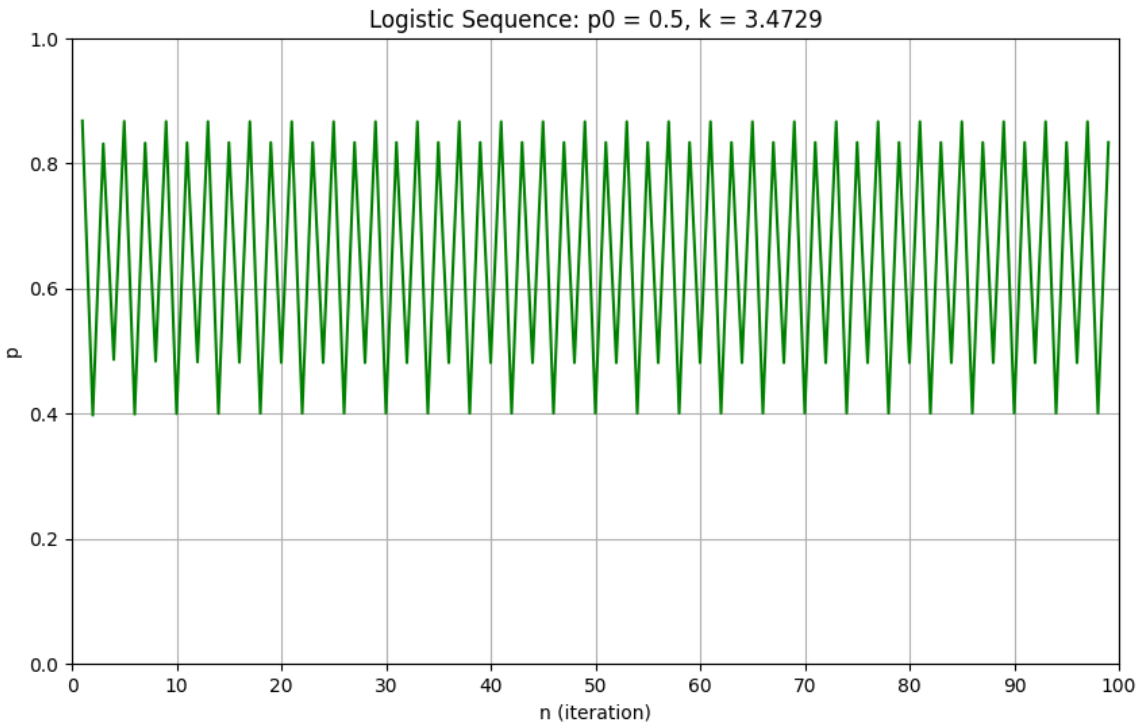


Experiment2 - Observations

- This type of behavior, where the sequence enters a fixed pattern of repeated values, is known as a **periodic oscillation**. It indicates that while the graph does not settle to a single point (as in the convergent case), it stabilizes into a cycle. The number of distinct values in this cycle (its period) depends on both k and p_0.

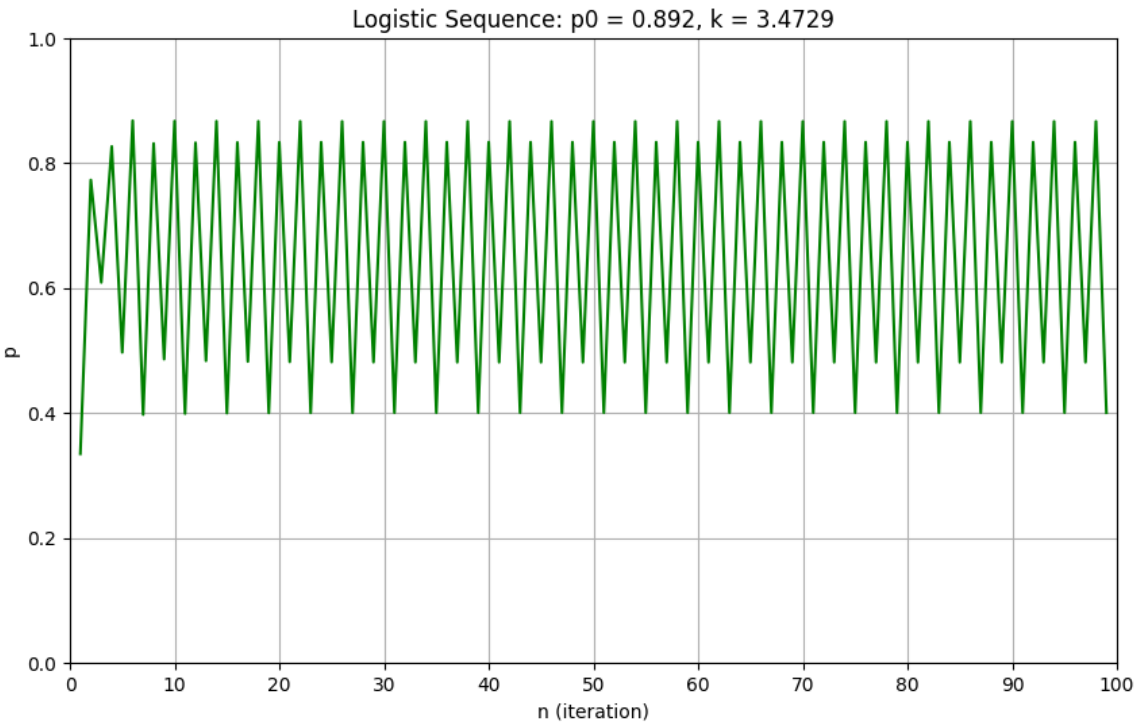
Experiment3: $3.4 < k < 3.5$

- We selected k = 3.4729 and observed the following:
- For $p_0 = 0.5$, the sequence exhibits periodic behavior, but the oscillations are uneven. The values fluctuate irregularly between approximately 0.4 and 0.9, with alternating highs and lows that do not stabilize to two fixed values like in previous cases.



Experiment3: $3.4 < k < 3.5$

- We selected k = 3.4729 and observed the following:
- For $p_0 = 0.892$, the pattern is similar. After a few iterations, the sequence enters a periodic cycle, but the oscillations vary in size, ranging from around 0.5 to 0.9, without settling into a simple repeatable loop.

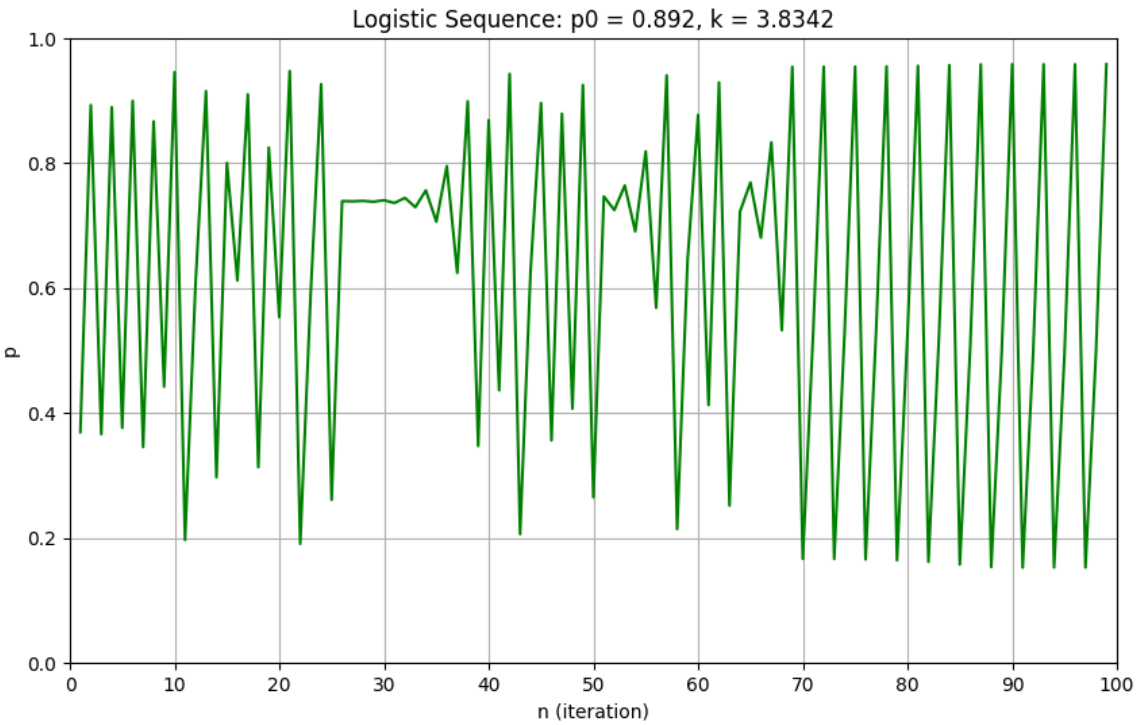


Experiment3 - Observations

- This indicates the presence of **complex periodic behavior**, where the system cycles through multiple values in an uneven pattern. It does not converge to a stable limit or a simple 2-point oscillation, suggesting a more complex structure.

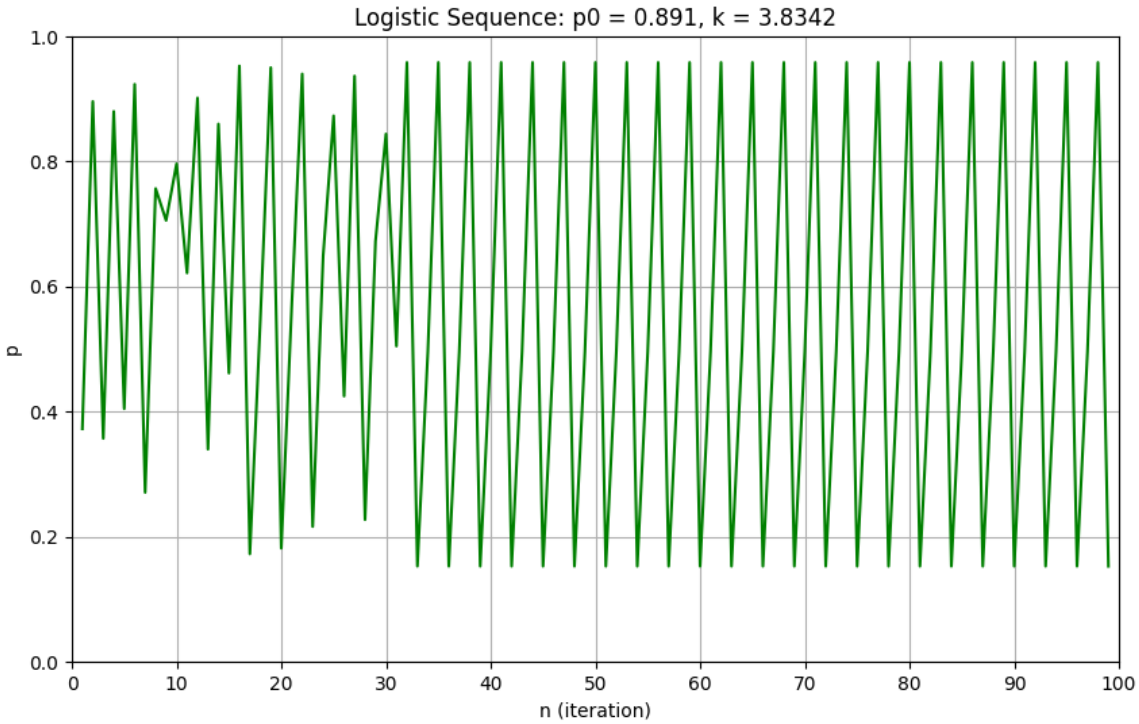
Experiment4: $3.6 < k < 4$

- We chose $k = 3.8342$
- For $p_0 = 0.892$, the graph is highly chaotic. The sequence displays irregular oscillations with no clear pattern: sharp dips, rapid rises, periods of flatness, and segments of regular oscillation all appear within the same sequence.



Experiment4: $3.6 < k < 4$

- We chose $k = 3.8342$
- For $p_0 = 0.891$, despite the small change of only 0.001, the resulting sequence differs significantly. It too is chaotic, but the exact pattern of ups, downs, and plateaus is distinct from the first case. This highlights a feature of this behavior: extreme sensitivity to initial parameters.



Experiment4 - Observations

- These results illustrate what is commonly referred to as **chaotic behavior**. In this context, prediction becomes kind of impossible due to how drastically tiny changes in p_0 can affect the outcome. Such results are known to appear in real-world models, such as insect population models under certain conditions.

Conclusion

- Through a couple of experiments and graphs, we explored how the logistic sequence behaves under different values of the growth rate k and initial population p_0. We observed a range of behaviors: convergent, periodic, and chaotic; each revealing unique characteristics into population dynamics. The experiments highlight importance of this simple model, particularly its sensitivity to initial conditions in the chaotic outcome. This reinforces how even basic recursive functions can produce complex and unpredictable outcomes.

Sources

- *Calculus* by James Stewart (for recurrence relations)
- Milefoot.com: Logistic Sequences
- Wikipedia: Logistic Map

Thank you!

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